

LEO MURPHY

Mechanical & Automation Engineer

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ENGINEERING EXPERIENCE

Automation Engineer, Murphy International

March 2024 – December 2024

Worked full-time in the automation department on industrial control systems. Responsibilities included PLC programming, designing and wiring control panels, sourcing components, and producing technical documentation for installed systems.

SolidWorks Demonstrator, Dublin City University

September 2024 – November 2024

Delivered hands-on tutorials to first-year engineering students, guiding them through core SolidWorks functionality, including sketching, part modelling, and assemblies.

EDUCATION

Integrated BEng/MEng, Mechanical and Manufacturing Engineering with Simulation, Dublin City University

September 2021 – May 2026

Completed a four-year integrated programme combining a BEng in Mechanical and Manufacturing Engineering with an MEng specialisation in Mechanical Simulation. Developed a structured, project-based approach to engineering problem-solving through major design builds, including the Final Year Project, the Fireman robot, and the Autonomous Guided Vehicle (detailed below). Specialised in CFD and FEA using ANSYS to simulate mechanical responses and fluid-flow behaviour, and completed an independent research project requiring a critical review of academic literature.

SKILLS

- PLC programming and industrial control systems
- CAD design — SolidWorks (CSWA, CSWP Segment 1)
- CFD and FEA simulation (ANSYS)
- Embedded systems and microcontroller programming (Arduino, ATmega)
- Strong mathematical and analytical problem-solving
- Microsoft Excel — formulas, data analysis, charting
- Effective collaborator in team-based engineering projects

PROJECTS

Final Year Project — Automated DNA Amplification Platform

Designed and built an automated platform for biological assay procedures (LAMP and PCR DNA amplification). A primary motor spins an offset disk, generating centrifugal forces to move liquids; a secondary motor reorients the force vector under closed-loop PID control. An integrated heater drives the amplification reaction, while a camera detects colour changes and reports results to the master node. The master node manages disk inventory, instruction sequencing, inter-node communication, and timing.

Robot Arm (ongoing)

Designing and building a multi-axis robot arm. Built an initial small-scale prototype to validate inverse kinematics calculations, followed by a second prototype with simple geared axes to test stepper-motor control using magnetic encoders. The next iteration will be a 5-axis arm using NEMA 17 steppers, geared down via cycloidal drives at the base and lower-arm joint, with a capstan drive for the upper arm.

PCB Business Card

Designed a custom PCB business card powered by a 3.3V coin cell, featuring five buttons and a charlieplexed LED display capable of running simple games. Built around an ATmega328P-AU microcontroller.

Control Panels (Murphy International)

Designed and built a range of demonstration control panels, including a school outreach unit featuring motor control, an HMI, traffic-light logic, and interactive games to introduce ladder logic and PLCs to secondary-school students.

Fireman Project

Designed and built a small robot to climb a 1m vertical ladder while carrying a beaker of water without spillage, then raced it head-to-head against competing designs. The challenge demanded creative problem-solving within tight constraints on materials, components, and manufacturing processes.

Heart-Rate Sensor Proof of Concept (2022)

Built a proof-of-concept optical heart-rate sensor using an LDR, green LED, and Arduino. The device illuminates the skin and measures voltage variations across the LDR, applying a 5-point moving average to smooth the signal. Successfully detected pulse peaks corresponding to blood-flow changes; early iterations did not compensate for ambient light, introducing baseline drift. SpO₂ measurement attempts were inconclusive but provided a strong basis for further iteration.

Boat Project (2021)

Designed a small boat in SolidWorks and analytically derived its velocity-time curve, tilt angle, and travel time over a fixed distance. 3D-printed the design and tested it experimentally to validate the analytical model.

Autonomous Guided Vehicle (2021)

Designed and built a small vehicle to autonomously follow a line, perform a three-point turn, and trace its route in reverse — all within strict material and dimensional constraints.

Aluminium Casting (2020)

Built a small home foundry from a repurposed Hoover and 3D-printed components. Used 3D-printed pattern blanks to create sand moulds for casting aluminium parts.

OTHER EXPERIENCE

Farm Labourer, Family Farm

2014 – Present

Contribute to the daily running of the family farm across mornings, evenings, and weekends. Responsibilities include feeding, dosing, herding, and treating livestock, alongside general maintenance. Routinely operate heavy machinery including telehandlers and tractors.

Labourer (Formworker), AK Civil and Construction

June 2023 – August 2023

Worked full-time as a formworker on civil construction sites. Built shutters, tied rebar, and finished concrete pours, working both independently and as part of a small crew.

Warehouse Picker, Musgraves

May 2022 – September 2022

Worked full-time in a high-volume warehouse, picking orders to tight deadlines as part of an integrated team. Operated pallet trucks and other handling equipment under strict health and safety procedures.

CERTIFICATIONS

- Certified SolidWorks Associate (CSWA) — 2022
- Certified SolidWorks Professional, Segment 1 (CSWP) — 2022
- Safe Pass

REFERENCES

References available on request.